Assignment 1: Effect of Marketing Spend on Purchase Amount

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Introduction

Over the last quarter, our company ran a marketing campaign where we varied how much we spent contacting individual customers (ads, emails, coupons). We will be using this data to determine if higher marketing spend per customer lead to higher purchase amounts. We would also be comparing the data between different customer segments.

Research Questions

Using the data from the campaign, this report investigates whether higher per-customer marketing spend is associated with higher purchase amounts. The aim is to assess whether increasing marketing investment is an effective strategy for driving customer spending and to gain insight into the potential return on investment.

Dataset Introduction

The dataset consists of customer-level observations from a marketing campaign carried out over a quarter by a retailer. Each row represents a unique customer capturing both the marketing spend allocated to them and their resulting purchase behavior.

Data Cleaning

```
spc_tbl_ [150 × 7] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ customer_id    : chr [1:150] "36621617_c0001" "36621617_c0002" "36621617_c0003"
"36621617_c0004" ...
 $ signup_date    : chr [1:150] "2025-09-05" "2025-01-27" "2025-03-07" "2025-04-04"
...
 $ marketing_spend: num [1:150] 250.8 215.2 47.1 190.1 184.8 ...
 $ visits         : num [1:150] 9 8 2 10 8 5 9 7 2 4 ...
 $ purchase_amount: num [1:150] 290.4 127 71.2 60.5 186.6 ...
 $ segment        : chr [1:150] "Consumer" "SMB" "SMB" "SMB" ...
 $ notes          : chr [1:150] NA NA "no response" NA ...
 - attr(*, "spec")=
  .. cols(
  ..   customer_id = col_character(),
  ..   signup_date = col_character(),
  ..   marketing_spend = col_double(),
  ..   visits = col_double(),
  ..   purchase_amount = col_double(),
  ..   segment = col_character(),
  ..   notes = col_character()
```

```
  .. )
 - attr(*, "problems")=<externalptr>
```

The dataset contains 150 observations and 7 variables. Initial inspection of the data structure revealed several issues that required cleaning prior to analysis.

Firstly, sign_up variable was stored as a character string in multiple inconsistent formats. To ensure consistency, it was converted into a standard date format using the ‘parse_date_time()’ function.

Secondly, the dataset contained missing values across several variables. Observations with incomplete values were removed to ensure the reliability of the analysis.

Finally, the notes variable was excluded from the dataset, as it contained non-essential information that was not relevant to the research objectives.

Dataset Description

After cleaning, the data set has 124 rows and 6 columns.

Campaign Summary

Customer Segment	Total Customers	Average Spend	Average Purchase	Average Visits
Consumer	70	165.91	132.46	5.14
Enterprise	13	160.13	149.35	6.08
SMB	41	154.21	121.70	5.93

The above table gives us a summary of how many observations have been recorded for different segments, along with the average spend, purchase and visits for each of the segments.

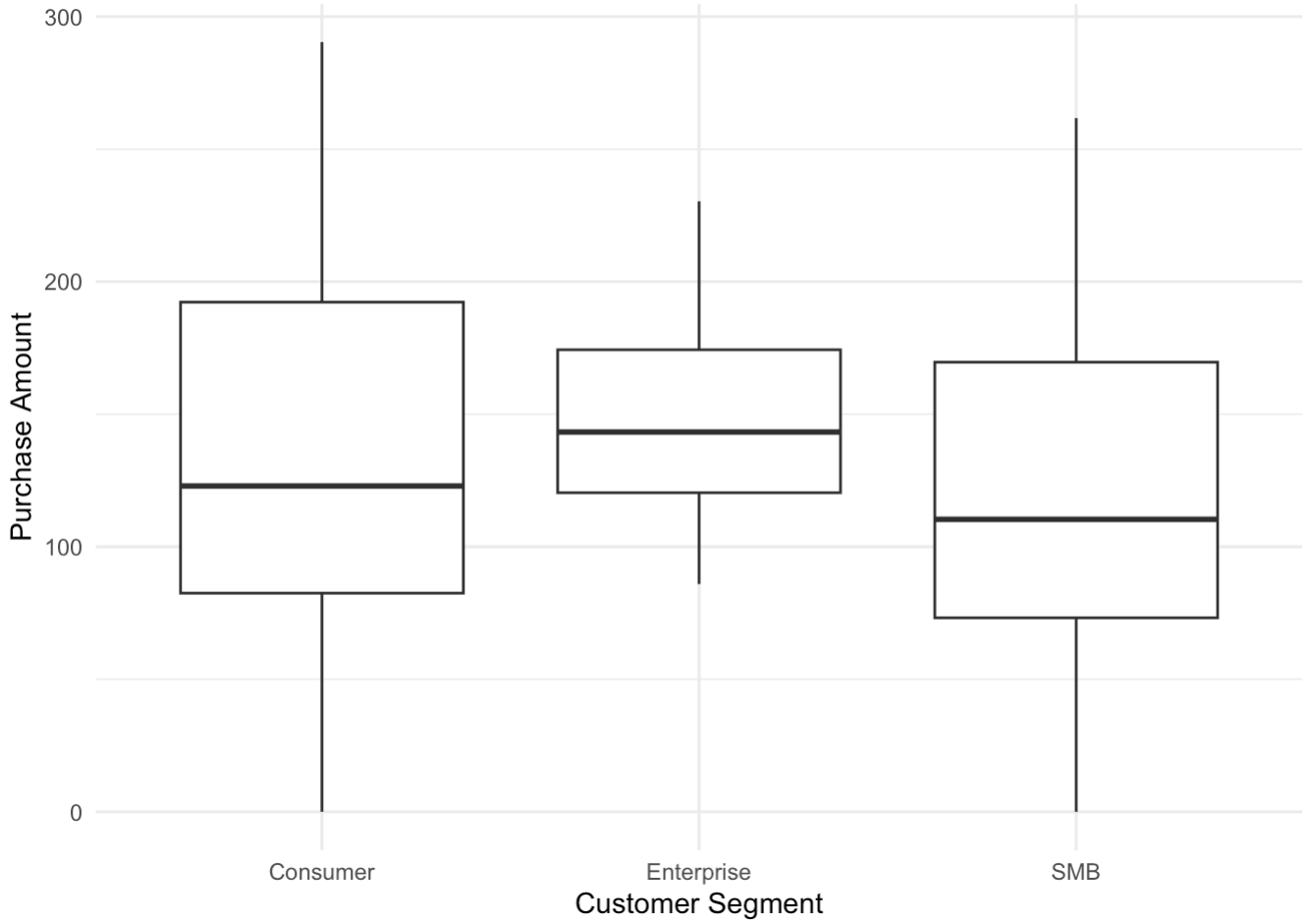

Figure 1: Distribution of purchase amounts across customer segments

The distribution of purchase amounts varies across customer segments, as shown in Figure 1. Looking at the box plot, we notice that enterprise customers have the highest median purchase amount. In contrast, SMB customers have the lowest median purchase amount and also show a wider spread suggesting greater variability in spending behavior. Consumer customers fall between the two.

Results

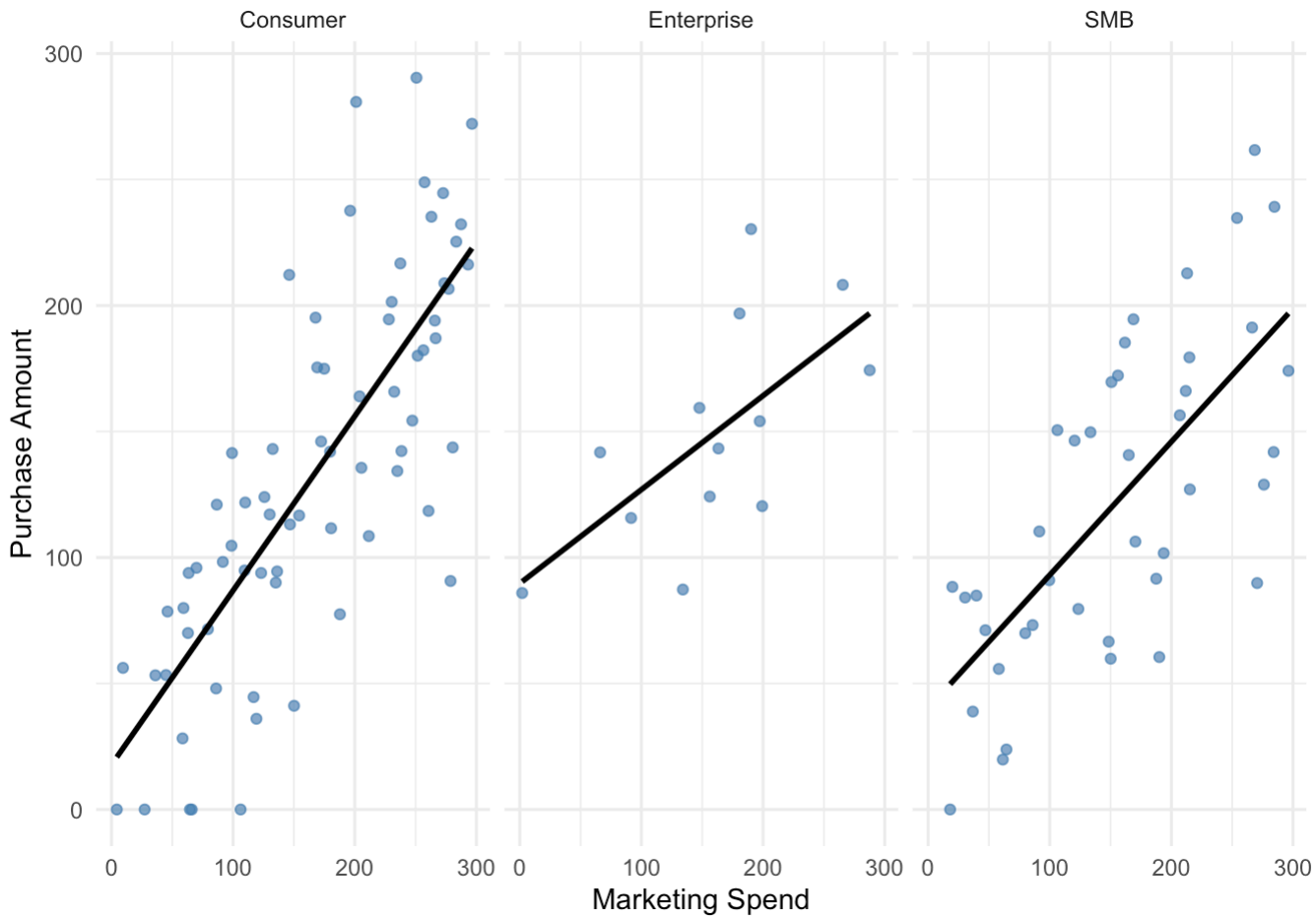

Figure 2: Relationship between marketing spend and purchase amount across customer segments

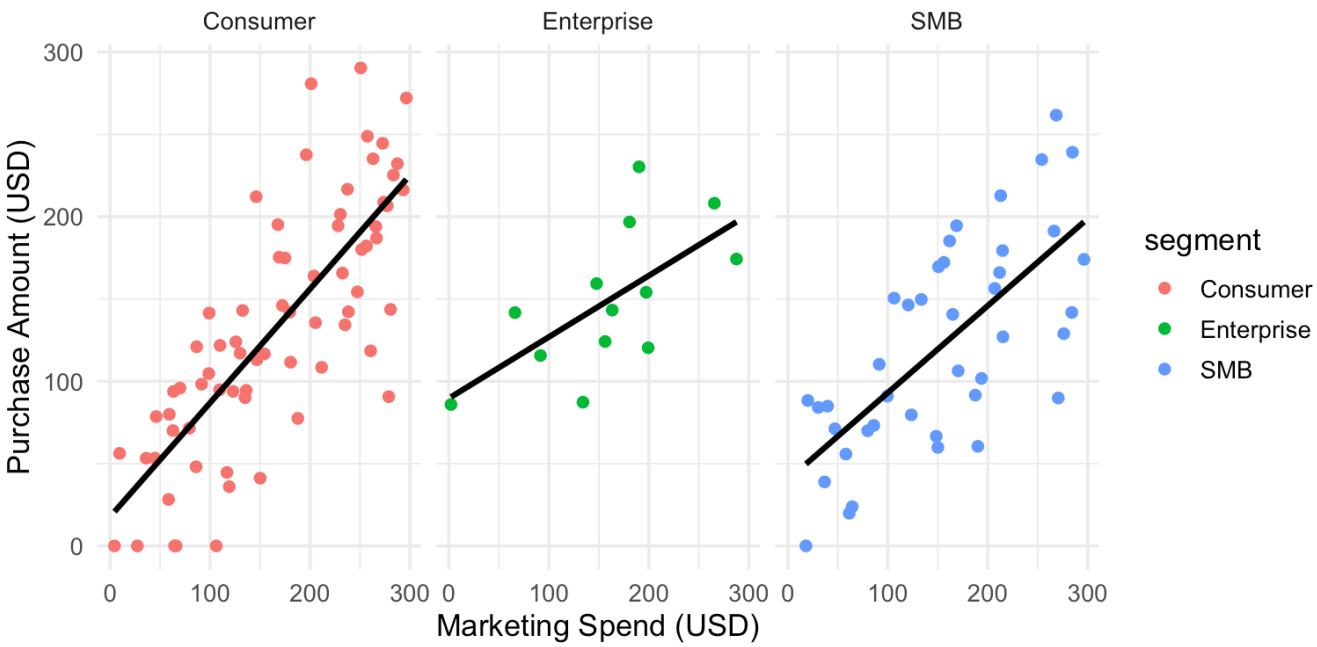

Figure 3: Scatter plots of marketing spend versus purchase amount, split by customer segment

```
Call:
lm(formula = purchase_amount ~ marketing_spend, data = cleaned_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-111.695  -33.447    0.807   33.228  125.835 
```

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    32.04578    9.07601   3.531 0.000585 ***
marketing_spend  0.61096    0.05012  12.190  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 45.79 on 122 degrees of freedom
Multiple R-squared:  0.5491,    Adjusted R-squared:  0.5454
F-statistic: 148.6 on 1 and 122 DF,  p-value: < 2.2e-16
```

The relationship between marketing spend and purchase amount was explored using a scatter plot and a linear regression model. As shown in Figure 2 there is a positive relationship between the two variables. This indicates customers who receive higher marketing spend tend to spend more.

A linear regression model was fitted to measure this relationship. The results show that marketing spend is a good predictor of purchase amount ($p < 0.001$). The estimated effect is 0.611, meaning that for every 1 unit increase in marketing spend, purchase amount increases by about 0.611 units on average.

The R-squared value is 0.549, which means that about 54.9% of the variation in purchase amount can be explained by marketing spend. This suggests that marketing spend has an important but not complete influence on customer purchases.